

ISSB Limited

Commercial Systems Development Plan

Produced by Charles Jardine

July 2019

Table of Contents

1.0 Introduction and Summary	4
2.0 Background	4
3.0 Company Direction	5
3.1 Delivering business continuity by moving to the Cloud	5
3.2 Reduce dependency on the current on premise servers.....	5
3.3 Automation of Data Loading.....	5
4.0 Operations Review and Solutions.....	6
4.1 Issues Arising.....	6
5.0 Security and Authentication.....	6
6.0 CRM System.....	8
7.0 Automation of Data Uploader and importing	8
7.1 Data Storage and retrieval	10
8.0 Conclusion and Potential Solutions	11
8.1 Current Risks	11
9.0 Software Functional Specification.	12
9.1 Authentication	13
9.2 CRM System	14
9.3 Data Importing	15
9.4 Data Updates	16
9.5 Customer Self Service and reporting.....	17
10.0 Server Commissioning.....	18

11.0 Application Bootstrap	18
12.0 Data Migration	18
13.0 Maintain static data	19
14.0 Help Facility	19
15.0 Archive / Delete Facilities.....	19
16.0 Audit transactional events	19
17.0 Record Stamping	19
18.0 Application Interfaces	20
19.0 Application Re-Branding	20
20.0 UK using Intrastat data feed and reporting	20
21.0 Data Feeds from 3rd party sources	20
22.0 Development Time Scales	21

1.0 Introduction and Summary

This document reviews the systems support at ISSB Ltd and proposes future developments to improve the efficiency and effectiveness of business processes.

2.0 Background

The International Steel Statistics Bureau offers a wide range of publications and custom reports covering UK, European and Global trade in steel and raw materials. The company maintains a comprehensive database of international trade in steel and steelmaking products containing approximately 550 million records accumulated since 1980. ISSB data is used by industry suppliers, consumers, consultancies, financial and investment businesses, shipping companies, media companies and the European commission and Governments.

ISSB is currently running a number of legacy systems that demand a significant amount of manual intervention to manage the data and generate the information requested by their customers. This includes the Trade enquiry system, bespoke CRM tool, and Access client data loader, IBM Redbrick server with data stored in Microsoft SQL Server and business objects which are all hosted within the ISSB offices.

The proposed Open Data Platform will replace all existing legacy hardware and licensed software components in ISSB with a single cloud-based platform that will automate a number of activities and improve the customer experience.

Therefore, Nathan Software have been asked to review the operating processes and support applications and define a functional specification for improvements.

This document is the result of that review.

3.0 Company Direction

Key objectives to support business growth

- Providing 24x7 access to data
- Improving efficiency through automation of data management
- Increasing sales by delivering a better experience for customers
- Server Technology pushed to the cloud reducing the risk of failure

3.1 Delivering business continuity by moving to the Cloud

ISSB currently host their data services on site. As well as requiring constant maintenance: component replacements, upgrades, damage protection, location security, and local network troubleshooting and staff system administration on-premise environments are more susceptible to incidents that may limit access to the data. As a data organisation ISSB is dependent on ensuring that their customers can access data on demand and in the past ISSB have suffered an incident that has meant that data has been unavailable for several days.

As a Cloud solution the Open Data Platform will offer ISSB a scalable managed infrastructure tailored specifically to the company's demands, improving efficiency and delivering business continuity. ISSB can focus on its customers knowing that the data is available.

3.2 Reduce dependency on the current on premise servers

Moving to the cloud will remove the need for ISSB to host and manage their own servers and the complexities involved in this around security, patching, upgrading and redundancy. Current licensed software products such as Microsoft SQL will not be required. Automatic backups will be done in the Cloud removing the risk of potential disaster.

3.3 Automation of Data Loading

ISSB receive their data in a variety of formats, a bespoke Microsoft Access client tool is used to load the data into their IBM redbrick database. The tool was developed for Access 2003 and it does not work on the latest versions of Access, with no potential way to migrate it to the latest version of Access. Microsoft offer no support for Access 2003 and is unlikely to run on newer Microsoft Operating Systems. ISSB are currently running OS v7. The latest version is OS v10.

The Open Data Platform will remove the need for the Access client data loader, the platform will enable the automation of loading data by providing a file drop location for the data file to be loaded, it will then extract the data into a temp location and map to the data store when the mapping is complete it will load the data into the platform and make it available for searching. If there are any errors in the data file, it will notify the data provider and the amended data will be loaded.

4.0 Operations Review and Solutions

4.1 Issues Arising

ISSB Ltd have been successful over a 20-year period when informal systems and processes could satisfy current levels of demand. However, as time goes on systems have become obsolete and unsupported, thus running a risk of failure.

These issues, listed below, are concerned with the day to day operation of the company and the support offered by Microsoft and IBM Ongoing. However, they must be viewed in the context that the current software was written for a different era.

5.0 Security and Authentication

	Problem	Result	Solution
1	The system must be secure and locked down using SSL certification	With no encryption data from client computers will be unencrypted and servers compromised	Apply for an SSL certificate from either and automated self-sign or use a global certificate like Comodo
2	At the moment the TES (Trade Enquiry System) system has single factor authentication	The user credentials can be passed around allowing multiple people to be logged in at the same time	Use 2 factor authentication, this will limit access to the system as a mobile phone will be required to log in
3	Each Account will have various user roles and profile based on country and allowable content	The system will lock down authenticated users to what they are allowed to access	Provide user access roles: Admin: ISSB employee access for administration User: with access to what he has paid for Demo: a limited view for sales purposes only

	Problem	Result	Solution
4	Account control access		Each user account will be controlled by an expiry date, on hold flag and locked flag
5	Accounts will be subject to allowable countries	Will only view allowable countries	ISSB will setup the countries based on an agreed contract
6	Accounts will be subject to allowable products	Will only view allowable products	ISSB will setup the products based on an agreed contract
7	Account will be subject to allowable date range	Will only view by allowable date range	ISSB will setup the date range based on an agreed contract
7	Each account user should have control of their password	The ability to change own password	Allow the user to change their password via the web portal
8	Each account user should have control of forgotten passwords	User updated password	Allow the user to be able to have a password reset link sent to their email
9	Each account user should have a profile page	User updated profile	The user can view and change his profile, for example display name

6.0 CRM System

	Problem	Result	Solution
10	The ability to hold customer details:	User data can be retrieved	To maintain and manage user accounts within the context of the CRM system. The data from the current CRM system will be migrated across
11	Monitoring system usage	The ability via KPI and dashboards view each user's usage of the system	Build activity logs of user activity, countries, products, ext.
12	Sales opportunities are being lost, with the implementation of item 11 this will enhance the process	Lack of revenue	Closely monitor the user's activity with a view to upsell

7.0 Automation of Data Uploader and importing

	Problem	Result	Solution
13	The current system receives emails or downloads from various data providers. The data has to be transferred, in many cases, transformed into a format recognisable by the server manually by ISSB Staff	Inefficient use of manpower and the possibility of introducing error into the system	The data files can be manually uploaded into the server via the portal where validation is completed within the server followed by a data import
14	The current system receives emails or downloads from various data providers. The data has to be transferred, in many cases, transformed into a format recognisable by the server manually by ISSB Staff	Inefficient use of manpower and the possibility of introducing error into the system	The data files will be sent to a dedicated DATA mailbox where the server will constantly poll for new data files. The server will automatically retrieve the import files from the mail box, validate the data and import the data

	Problem	Result	Solution
15	Data import logs are required to record data as it arrives into the system	A record of each piece of country data will be logged	Build a logger system that will trigger an email to subscribed Users that a new country has been added. The log will also be used to remove data that may have been provided incorrectly
16	Alert the User of new imported data	The User will receive an email alerting them on new country data added to the system	Build an alert system for users if they want alert notifications, this will be setup within the user profile
17	Port Data required. The current infrastructure does not support port data that is supplied within some data files	Insufficient information provided to the user	Capture and analyse the port data for each country when available
18	With the methods used today to get data into the system results is a high dependency on staff knowledge and skills.	Higher operating costs	Provide better systems and information on the computer that will alleviate the dependency on the memories of office staff.

7.1 Data Storage and retrieval

	Problem	Result	Solution
19	Currently the Data is split between 3 Database offerings, IBM Redbrick, MS SQL Server and MS Access, this is costly and prone to data error being introduced.	Higher maintenance, lots of backups with higher operating costs	Using 1 database, MongoDB document type database, placing all data into 1 repository and enhancing speed of searching and retrieval
20	Business objects, the current system uses an out of date set of Business Objects tools extensively, creating client bespoke reports	Replacement costs are high and high operating costs	Provide a user set of tools on the portal, the user will have the ability to design and run their own reports on demand. This will eliminate the need to have Business Objects, reducing high operating costs and giving more control to the user

8.0 Conclusion and Potential Solutions

From the evidence presented above it is obvious that the systems are out dated, not supported by Microsoft and runs the risk of down time and as mentioned before in our proposal:

8.1 Current Risks

- Dependency onsite servers that if disaster occurs then difficult to plan for contingency
- Ageing technology stack with no easy route to upgrade
- Multiple manual activities, heavily depending on existing staff to execute.
- Customer experience is not as joined up in the process, potential to for revenue earning opportunities to be missed.
- License costs and upgrading of existing platforms.
- Backups are manual and stored in a safe on-site

9.0 Software Functional Specification.

The remainder of this document is based on the bespoke solution for the commercial aspects of ISSB Ltd.

This section of the report describes the business processes and how they should be supported by information technology systems. The processes are described in the format below.

Business Process <i>Description of what the business process and how it works</i>	Trigger Inputs Outputs Rules Data Created Data Updated Data Used Frequency Mode	<i>What starts the business process</i> <i>What goes into the process</i> <i>What results from the process</i> <i>Business rules governing the process</i> <i>What data is created on the successful completion of the process</i> <i>What data is updated on the successful completion of the process</i> <i>What data is used by the process</i> <i>How often does the process take place</i> <i>How does the process trigger start</i>
System Requirements	<i>Description of what the system should do to assist the process</i>	

9.1 Authentication

User Login User is logged into the system	Trigger Inputs Outputs Rules Data Created Data Updated Data Used Frequency Mode	User keys in his credentials User Name, Password, 2 Factor Auth code User portal is presented to user All authenticated users must have a contract with ISSB, be in date and fully paid up subscriber A log of the user will be recorded when created A log of the user will be recorded when updated A log of the user will be recorded when used Allow access any time Internet Portal access
System Requirements	The system will use 2 factor authentications however this can be turned off depending on the role of the user. For example, this would not be required for ISSB staff or administrators. Each user will have to be authorised by ISSB before system access was allowed: 1. A contract has been agreed between the 2 parties 2. Expiration date has been set for the contract 3. Disable the lock flag to allow access 4. Setup allowable countries 5. Setup allowable products 6. Setup allowable year range for reporting, i.e. 2008-2019 7. The user has the ability to change their password 8. The user has the ability to retrieve their password	

9.2 CRM System

Receive User Details Process data for the User. Receive information and verify a customer trading agreement exists or not. Enter the user's details into the CRM system	Trigger Inputs Outputs Rules Data Created Data Updated Data Used Frequency Mode	User sends order, or signed up to ISSB portal User details Acknowledgement / confirmation that ISSB has received their information Only valid users who do not have an embargo placed upon them can be supplied. If the user is subject to a temporary embargo his payment must be cleared before access to the system is allowed. User must have sufficient credit to access the system. CRM Data record created When Changes occur for the user Communications data recorded Mail, e-mail, fax input, e-commerce and on-line processing
System Requirements	The system will provide the ability to record, delete and change user information, banking and financial information and transactions of any activity on the system. The system will present KPIs on each user and collective users via data cells and dashboard integration. Monitor and capture Geo Location data based on IP address information. Record information each time the user contacts ISSB or ISSB contacts the user	

9.3 Data Importing

System Receives data Process data for the system	Trigger	Data is received into the data exchange email account or is uploaded via the web portal
	Inputs	File is detected automatically via email scan or upload
	Outputs	The data is validated against the appropriate country profile and entered into the main data repository
	Rules	The data received into the system must have the following attributes: 1. Has a valid country GEO import and export Code 2. Has a valid tariff code 3. Trade flow indicator is valid 4. Processing date is valid, i.e. does it already exist 5. If Port Data? Then validate 6. Apply currency rules where applicable 7. The data must have a method of delete and rollback
	Data Created Data Updated Data Used Frequency Mode	Record in a log file of each data insertion header Record in log file Record in User Log Record in log file
System Requirements		The system will import data into MongoDB based on email scan or portal uploader. The system will monitor, and record event logs each time country data is entered into the system. For users that have applied a notification on their profile to be notified when data arrives, a notification will be sent via email or text via a text gateway.

9.4 Data Updates

System Receives data change	Trigger	Data is changed in the portal
Processed data change events in the system	Inputs Outputs	When data changes have been changed, updated or deleted A log file will record each change and who changed it
	Rules	The system will maintain an audit log of data changes made to the system. 1. Each change will be logged by date, type and by user 2. Product tariff code updates, insertions and deletions 3. Country updates, insertions and deletions 4. User updates, insertions and deletions 5. User updates, insertions and deletions 6. Currency updates, insertions and deletions 7. User reports and search data 8. Changes can be undone from the log 9. Data validation rules will be checked based on 9.3
	Data Created Data Updated Data Used Frequency Mode	Record in a log file Record in log file Record in log file Record in log file
System Requirements	The system will maintain an audit log for each change that occurred in the system, each change can be undone and restored and flagged by date and user. Only users that have administrator roles to the system can make these changes.	

9.5 Customer Self Service and reporting

User Self Service portal	Trigger	User requires data from the system
Running User own reporting system	Inputs	The user will choose from an originally created report or create a new report
	Outputs	Data will be viewable via table views that can then be generated into graphs of various types
	Rules	The system will give the user access only to data that they have signed up for, the user will data select on: 1. Primary Country data by range and grouping 2. Secondary Country data by range and grouping 3. Product level selection by range and grouping 4. Trade Flow Selection 5. Ports if available 6. Value and tonnes 7. Date range by quarter, month, year and grouping 8. Grouped by any of the above
	Data Created	Users own defined data reports
	Data Updated	Users own defined data saved into system
	Data Used	Recorded to log
	Frequency	Recorded to log
	Mode	
System Requirements	The system allows the user to create and store reports based on their selection, these reports will be stored as live reports viewable within the context of a dashboard, graphs, and downloadable table views. Visualisation and dynamic data updates on login.	

10.0 Server Commissioning

One of the primary objectives that is fundamentally key to this project is commissioning a hosting company that will support the system for the next 5 years and beyond. I have recommended a fixed cost solution using Fast Hosts. Microsoft Azure or Amazon Cloud services are unknown costs or pay as you go solutions that ultimately could cost ISSB substantially more month on month.

Microsoft Azure do offer more facilities, but costs may be prohibitive.

No work can be started until the hosted solution is in place.

11.0 Application Bootstrap

The application template has to be established before coding can be started.

The following will be established:

1. Setting up the development platform in Azure Dev Ops
2. Setting up the project template using DOT NET CORE
3. Design using CSS and jQuery
4. Build code...

12.0 Data Migration

Data already being used in the current live system has to be re-engineered and migrated over to the new MongoDB. This will take a considerable amount of time.

1. Data Applications will be written to do the Migration
2. Data extraction from Access where applicable
3. Data extraction from MS SQL server
4. Data extraction from Redbrick database
5. Data after migration will be validated
6. Data will be re-engineered to perform better and for querying

13.0 Maintain static data

The system should provide facilities to create, update and delete static data whilst preserving the integrity of the data. That is, it must not delete a parent while there are dependent children such as deleting a Product Group while Products exist for it. It should also provide basic reporting facilities for this data.

14.0 Help Facility

The system should provide an on-line Help facility for each screen and window. This Help facility will provide basic guidance on using the function, description of each data item displayed and a statement on the meaning and correction of any application generated errors. An index to the Help facilities should be provided enabling the user to select by a drop down menu the facility to be enquired upon.

15.0 Archive / Delete Facilities

Facilities should be provided to archive all the key transactional data after a period of time since they were active. This facility should maintain the integrity of the data model. A restore archived data facility should be provided to enable to user to view data which has been removed from the system because of its age. This facility will enable the management of the growth, in terms of volume, of data.

16.0 Audit transactional events

Facilities should be provided to audit the transactional data described. This facility should record a before and after update image of the data to be held. Basic reports showing the changes to this transactional data should be provided.

17.0 Record Stamping

The user name, date of update and date of creation should be held against each record on the database. Facilities should be provided to enquire upon this information. These facilities are outside the normal transactional processing.

18.0 Application Interfaces

The system will have the ability to interface with 3rd party companies via an API interfaces if required see 20,21

19.0 Application Re-Branding

The system can be re-branded and styled, and services sold on to a 3rd party provider, this will allow a 3rd party to host their own brand of the system, maintain their own CRM and use the core ISSB data set.

20.0 UK using Intrastat data feed and reporting

The system will allow development of a data feed from UK Intrastat, this is out with the scope of this document and requires further specification. Within Contract Cost.

21.0 Data Feeds from 3rd party sources

The system will allow development of a data feeds from 3rd party sources via an Application Interface, this is out with the scope of this document and requires further specification. Within Contract Cost.

22.0 Development Time Scales

The time involved in developing and implementing the solution described in this document has been detailed below. It is however recognised that the core project should be delivered within a six-month period.

Feature	Description	Duration
Server Commissioning	Core System	1 Week
Application Bootstrap	Core System	1 Week
User Login	Core System	2 Weeks
Data Migration	Core System	1 Month
CRM System	Core System	1 Month
Data Importing	Core System	1 Month
Data Updates	Core System	1 Month
Customer Self Service and reporting	Core System	1 Month
Maintain static data Help Facility Archive / Delete Facilities Audit transactional events Record Stamping		1 Month
Application Interfaces		1 Month
Application Re-Branding		1 Month